

Assignment 4: Regularization

Q1. Please answer the following questions in your own words.

1. What is the intuition of adding a penalty to mean squared error, that grows in the "size" (absolute or squared value) of the model parameters? Big coefficients usually mean the model is memorizing noise rather than learning real patterns. The penalty discourages this by making the model pay a price for every unit of coefficient size.
2. How does regularization provide a way of exploring the bias-variance trade-off? A higher penalty shrinks coefficients more, which makes the model have less variance but also more bias. You're basically tuning a knob between underfitting and overfitting.
3. What is the difference between LASSO and Ridge regression? How do the answers typically change for the two problems? Both shrink coefficients, but LASSO can shrink them all the way to zero which drops the variables entirely, while Ridge just makes them small. So LASSO does variable selection, Ridge keeps everything but smaller.
4. How do we typically scale variables for use in regularized regression? Why? If one variable is measured in dollars and another in years, their coefficients will naturally be very different sizes for unrelated reasons. Standardizing puts everything on the same scale so the penalty treats all variables fairly.
5. How is the penalty α typically selected? You try a bunch of values and pick the one that performs best on cross-validation. Bigger α means more shrinkage, smaller means less.
6. When conducting cross validation, do you include the penalty term in evaluating the cross validated MSE? Why or why not? No, you don't include it. You're trying to measure how well the model predicts new data, and the penalty is just a training tool so it has nothing to do with prediction accuracy.

Q2. This is a case study on regularization.

1. Import the `cars_hw.csv` dataset. Create an `Age` variable for each vehicle. Take `Mileage_Run` and `Age`, and (a) use `PolynomialFeatures` to create a third degree expansion, (b) use `StandardScaler` to z-score normalize them.
2. Use your features, run linear regression. What is the sign for the interaction between `Mileage_Run` and `Age`?
3. Use `LassoCV` to regularize your linear regression, using 20-fold cross

validation. (Hint: I used the grid `alphas = np.logspace(1,3,20)` to find the cost parameter)

4. Plot the cross-validated MSE by α .
5. Plot the coefficient paths by α .
6. Which features are actually selected? What proportion are set equal to zero?
7. Compare the linear regressions and optimally regularized coefficients. Do any coefficients increase in magnitude from linear regression to LASSO? Do any change sign?

```
In [ ]: import pandas as pd
import numpy as np
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import LinearRegression, LassoCV, Lasso

df = pd.read_csv('data/cars_hw.csv')

# 1. Create Age and define features
df['Age'] = 2026 - df['Make_Year']
X = df[['Mileage_Run', 'Age']]
y = df['Price']

# (a) Polynomial Expansion (Degree 3)
poly = PolynomialFeatures(degree=3, include_bias=False)
X_poly = poly.fit_transform(X)
feature_names = poly.get_feature_names_out(['Mileage_Run', 'Age'])

# (b) Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_poly)
```

```
In [4]: #2. Fit Linear Regression to the expanded and scaled features
from sklearn.linear_model import LinearRegression

lr = LinearRegression()
lr.fit(X_scaled, y)
coef_dict = dict(zip(feature_names, lr.coef_))
interaction_coef = coef_dict.get('Mileage_Run Age')

print(f"Interaction Coefficient (Mileage_Run * Age): {interaction_coef}")

if interaction_coef > 0:
    print("The sign of the interaction is Positive (+).")
else:
    print("The sign of the interaction is Negative (-).")
```

Interaction Coefficient (Mileage_Run * Age): -1671212.36
The sign of the interaction is Negative (-).

```
In [5]: # 3. LassoCV with 20-fold Cross Validation
alphas = np.logspace(1, 3, 20)
```

```
lasso_cv = LassoCV(alphas=alphas, cv=20)
lasso_cv.fit(X_scaled, y)

print(f"Optimal Alpha: {lasso_cv.alpha_:.2f}")
```

Optimal Alpha: 69.52

```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 13848094711.421875, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 546327167286.1875, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1462272063114.5703, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2032213950614.3281, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2336377869190.328, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2487472086649.875, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2883964983938.8516, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3002974401065.2266, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2846771195075.3125, tolerance: 12678950599.999998
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
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ou might want to increase the number of iterations. Duality gap: 268337
5918588.5625, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 257796
1121047.4766, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 252299
1971611.4062, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 243122
4013908.8438, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 406755
9932328.5703, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 368368
1149488.297, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 234216
1519618.3203, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 596429
5762447.141, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 108510
03227451.43, tolerance: 12678950599.999998
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 569510
865350.0312, tolerance: 12469725702.912619
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 140584
9280266.3047, tolerance: 12469725702.912619
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model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1892732013225.086, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2126897658627.3516, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2222830683712.6797, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2240198271908.633, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2718189415392.4844, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2693220961726.961, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3501022732276.883, tolerance: 12469725702.912619
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/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3392575661351.7734, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3350092761446.2344, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3231030505376.9844, tolerance: 12469725702.912619
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
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inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3942493203997.5938, tolerance: 12469725702.912619
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5233328276241.695, tolerance: 12469725702.912619
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6471517671379.75, tolerance: 12469725702.912619
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7655167880669.961, tolerance: 12469725702.912619
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8794885913718.773, tolerance: 12469725702.912619
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 587171062762.0156, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1292101523433.2734, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1764192098318.3203, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1986399432381.1797, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2628149691747.617, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 264182
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0964182.6406, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2496674216023.3516, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2340383090123.4453, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1934902065590.8984, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2187461589289.6016, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2442936052085.1016, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2545643172011.2812, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2568312568557.164, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3945762991193.383, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7783717532936.859, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12435326562197.023, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
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/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 16817960401941.418, tolerance: 12471993172.815535
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 32446254683.765625, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 837727963439.0156, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1568683667938.4219, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1964783314789.836, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2149333731151.9531, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2222968579676.7734, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2243949780946.7266, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2242504677804.9062, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2235046171795.758, tolerance: 12499790530.312838
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
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ou might want to increase the number of iterations. Duality gap: 281267
7397775.0156, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 331985
7557420.953, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 531953
6892219.406, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 508973
8920040.398, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 516869
3871062.1875, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 841875
2868504.211, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 126785
16603704.672, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 169008
27291895.32, tolerance: 12499790530.312838
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 517466
179452.2344, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 134001
1382045.7656, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 181297
6571028.1953, tolerance: 12160328699.02913
```

```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2031343705985.4375, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2112761643881.5078, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2129766452318.1875, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2120560692989.0, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2076903566708.8203, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2996014458846.6484, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5431015838569.258, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2877919679134.4688, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4466490583250.0625, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7774624335767.977, tolerance: 12160328699.02913
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11912384863814.86, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 15953311322962.523, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 19451794046784.312, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 22335448356460.426, tolerance: 12160328699.02913
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 17650776383.03125, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 545683296138.6719, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1402693072847.9219, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1903238851015.2812, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2144498530857.6016, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2244189401583.922, tolerance: 12832557326.213594
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 227488
```

```
4977286.039, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2275595933437.0938, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2543137212230.1016, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2709620303452.508, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2641883719075.3125, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4367971743504.297, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4904742637605.516, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3964612405074.5625, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7127153305068.766, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11553497351647.531, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 16041874932653.258, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
```



```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 19971176076333.37, tolerance: 12832557326.213594
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 19026104471.703125, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 480388309981.78125, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1248811910570.7266, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1678000981732.8516, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1870638900626.2812, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1940943566511.8203, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1955771762964.0156, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1948796715269.9766, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1934313059735.2422, tolerance: 12661919046.601942
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



```
ou might want to increase the number of iterations. Duality gap: 234525
1953804.875, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 238809
0446169.578, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 207126
2460629.6562, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 351772
5759443.8203, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 320511
1580488.9766, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 230163
8190687.086, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 605728
5875257.219, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 109731
54356066.125, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 156927
13964038.219, tolerance: 12661919046.601942
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 149816
33113.296875, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 427677
449762.71875, tolerance: 12836531798.058254
```

```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1624155438335.625, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2512832746100.8594, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2735540027717.992, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2679211587156.5625, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2563347186679.961, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2466782052326.539, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2402724152728.4688, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2363732047401.625, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2340580903186.6953, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2095786071100.9375, tolerance: 12836531798.058254
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3649603317517.828, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6175303027974.375, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9225856577537.703, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9545544833318.46, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10782751240413.969, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 13592112724586.188, tolerance: 12836531798.058254
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 392863441694.96875, tolerance: 12183129837.108953
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1271484992526.75, tolerance: 12183129837.108953
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1816543967983.5469, tolerance: 12183129837.108953
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2144990658836.586, tolerance: 12183129837.108953
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 281137
```

```
2604539.7656, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2812400136255.914, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2656489266265.0547, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2523638545929.8984, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2445236509646.1562, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2653728648170.8203, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2804173945903.4688, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4889364257290.125, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4462897021005.93, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4627898577220.164, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8135054689580.594, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
```



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/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12543799218810.023, tolerance: 12183129837.108953
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 100591883813.76562, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 981408190396.25, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1613317654801.4531, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1930977105433.1875, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2067557751762.0234, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2114761798665.3672, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2612010174341.039, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2628443369716.3906, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2486903912808.1953, tolerance: 12809397497.303125
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



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ou might want to increase the number of iterations. Duality gap: 265871
9836713.008, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 267817
9465082.547, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 476864
4439929.211, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 984674
1358617.414, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 619578
8186784.055, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 826249
7246968.336, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 119154
71614331.516, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 160467
52496600.219, tolerance: 12809397497.303125
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 209570
30978.125, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 401271
204641.3594, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 124415
5901442.4766, tolerance: 12604626778.856522
```

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model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1740576904864.4922, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1970954655010.9062, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2058703150988.75, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2305819113129.117, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2669793140582.1484, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2588085624611.953, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2446818122281.3594, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2351908628556.742, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2140512435869.4922, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3431014333422.7344, tolerance: 12604626778.856522
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4296958539146.3438, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2758774742963.9688, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5644101348184.758, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10089801303441.023, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 14756115475224.82, tolerance: 12604626778.856522
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 21719360166.59375, tolerance: 12304660007.98274
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 282043009879.5469, tolerance: 12304660007.98274
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1026826423296.5781, tolerance: 12304660007.98274
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1643634651941.4922, tolerance: 12304660007.98274
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1949528170077.9453, tolerance: 12304660007.98274
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 207657
```

```
0082430.6328, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2753638512750.2266, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2718622434919.4453, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2524097547590.672, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2685988288060.8984, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2537382185309.9844, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4547362752704.4375, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7344648815961.297, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9000844709058.64, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9324344383346.117, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11526058566257.07, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
```



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/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 14820190144687.195, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 18321230949456.445, tolerance: 12304660007.98274
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 303586239501.5625, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 887523333503.7656, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1396078087809.914, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1644302446329.6797, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2265749546412.8516, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2294521784670.3047, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2177502953677.3438, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2137430314708.6406, tolerance: 11826983510.895363
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



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ou might want to increase the number of iterations. Duality gap: 170431
8986992.289, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 184285
3809180.2969, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 210625
3057286.0625, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 227830
8830840.2188, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 487444
4465935.906, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 904027
0301285.336, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 134857
26102300.32, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 174293
95044035.125, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 206821
27143672.348, tolerance: 11826983510.895363
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 127564
439498.46875, tolerance: 10823131274.649408
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 110633
3685214.2031, tolerance: 10823131274.649408
```

```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2418382723998.836, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3698527498574.4297, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4858204439404.844, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5903528205553.977, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 689624834750.5859, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3447545884941.0234, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6280517513248.9375, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8708316757919.008, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10748737110829.023, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12499526955553.246, tolerance: 10823131274.649408
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 14049452510433.094, tolerance: 10823131274.649408
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 15462782609685.297, tolerance: 10823131274.649408
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 16781471178951.395, tolerance: 10823131274.649408
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 238265718165.09375, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1146322898973.1562, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1735746311711.4375, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2304162243918.508, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2839985273036.75, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2791815155488.7656, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2618950033648.4844, tolerance: 12733875574.325783
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 247991
```

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7921424.8906, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2397865620929.25, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2380785136802.633, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2440718845254.875, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4340328231963.6562, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7123118879070.875, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10282103223082.125, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9304422982358.414, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11573535090805.047, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 14998034491371.793, tolerance: 12733875574.325783
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 309528033359.53125, tolerance: 12847324822.22222
  model = cd_fast.enet_coordinate_descent_gram(
```



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/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1248809362107.8906, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1839110604525.9922, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2115378128934.2578, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2215363380001.0156, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2230526983264.992, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2632865173280.6484, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2611586748007.375, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2449207898552.6406, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2322381280128.1797, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2259130561117.25, tolerance: 12847324822.2222
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



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ou might want to increase the number of iterations. Duality gap: 230973
7335562.7188, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 405467
8801320.875, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 672678
7784225.445, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 741936
3369470.359, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 824413
4450469.664, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 108669
50081945.016, tolerance: 12847324822.22222
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 171645
84042.5, tolerance: 12932540441.379313
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 651145
965660.7344, tolerance: 12932540441.379313
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 146441
3200271.1719, tolerance: 12932540441.379313
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 191491
1501689.7031, tolerance: 12932540441.379313
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 212043
3817662.8203, tolerance: 12932540441.379313
```

```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2197182678706.711, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2214394080073.664, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2701499478596.7266, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2741981591393.086, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2598924959260.0547, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2475699452436.8438, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2407377902746.2734, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2378842543684.4297, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2283027003921.6875, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2004861555418.1172, tolerance: 12932540441.379313
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2329960085014.5938, tolerance: 12932540441.379313
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2596326344214.2344, tolerance: 12932540441.379313
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3136693812763.6094, tolerance: 12932540441.379313
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 269909010773.5625, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1203868254875.3906, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1808080456751.9297, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2100108896000.1172, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2712522010714.828, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2946368958069.164, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2813621964363.1562, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 264719
```

```
4531570.9766, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2596878151628.3125, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2599114755096.3047, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4629602080756.4375, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5959798610228.656, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5679514181196.398, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8015666672851.898, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11840274707509.633, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 16031274141222.457, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 19855867060963.49, tolerance: 12772170136.099136
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 444456040282.9844, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
```



```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1277585733369.5625, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1761883077676.5781, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2133399796013.6797, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2735193584760.3516, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2712367234192.242, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2551503611838.375, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2419800306072.4453, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2342620044547.8594, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2304448609484.758, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2287132360303.1562, tolerance: 12697436867.995687
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



```
ou might want to increase the number of iterations. Duality gap: 227905
5774901.633, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 227451
9138502.5547, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 227126
8173082.383, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 226860
4440333.125, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 226637
3234566.8125, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 226454
6500903.8125, tolerance: 12697436867.995687
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 108552
855614.89062, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 465665
847898.9531, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 115333
1644495.2969, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 173405
4304308.2266, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 201904
7820403.5469, tolerance: 12767023556.896553
```

```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2138253397602.75, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2176309911751.1562, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2178753544595.961, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2167973844201.7969, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2142666297836.6719, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3042170965635.9375, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4970752779952.523, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7671043129391.758, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10668275752959.39, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 13686457027690.46, tolerance: 12767023556.896553
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```

inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 13814629978807.062, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 15440945880393.395, tolerance: 12767023556.896553
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.236e+13, tolerance: 1.316e+10
    model = cd_fast.enet_coordinate_descent(

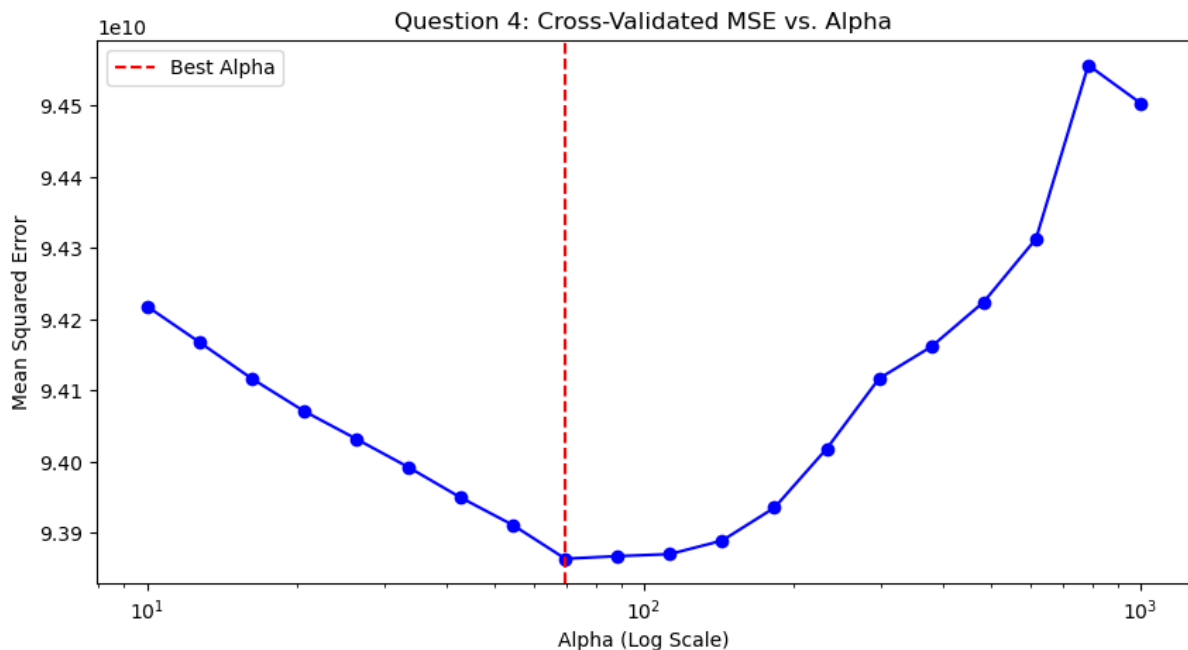
```

```

In [7]: # 4. Plot Cross-Validated MSE by Alpha
import matplotlib.pyplot as plt
mse_mean = np.mean(lasso_cv.mse_path_, axis=1)

plt.figure(figsize=(10, 5))
plt.plot(lasso_cv.alphas_, mse_mean, marker='o', color='blue')
plt.axvline(lasso_cv.alpha_, linestyle='--', color='red', label='Best')
plt.xscale('log')
plt.xlabel('Alpha (Log Scale)')
plt.ylabel('Mean Squared Error')
plt.title('Question 4: Cross-Validated MSE vs. Alpha')
plt.legend()
plt.show()

```



```

In [8]: # 5. Plot Coefficient Paths by Alpha
coefs = []
for a in alphas:

```

```

lasso = Lasso(alpha=a)
lasso.fit(X_scaled, y)
coefs.append(lasso.coef_)

plt.figure(figsize=(10, 5))
plt.plot(alphas, coefs)
plt.xscale('log')
plt.xlabel('Alpha (Log Scale)')
plt.ylabel('Coefficients')
plt.title('Question 5: Lasso Coefficient Paths')
plt.axvline(lasso_cv.alpha_, linestyle='--', color='black', alpha=0.5)
plt.show()

```

```

/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 4.197e+13, tolerance: 1.316e+10

```

```

    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 4.147e+13, tolerance: 1.316e+10

```

```

    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 4.085e+13, tolerance: 1.316e+10

```

```

    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 4.007e+13, tolerance: 1.316e+10

```

```

    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.909e+13, tolerance: 1.316e+10

```

```

    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.788e+13, tolerance: 1.316e+10

```

```

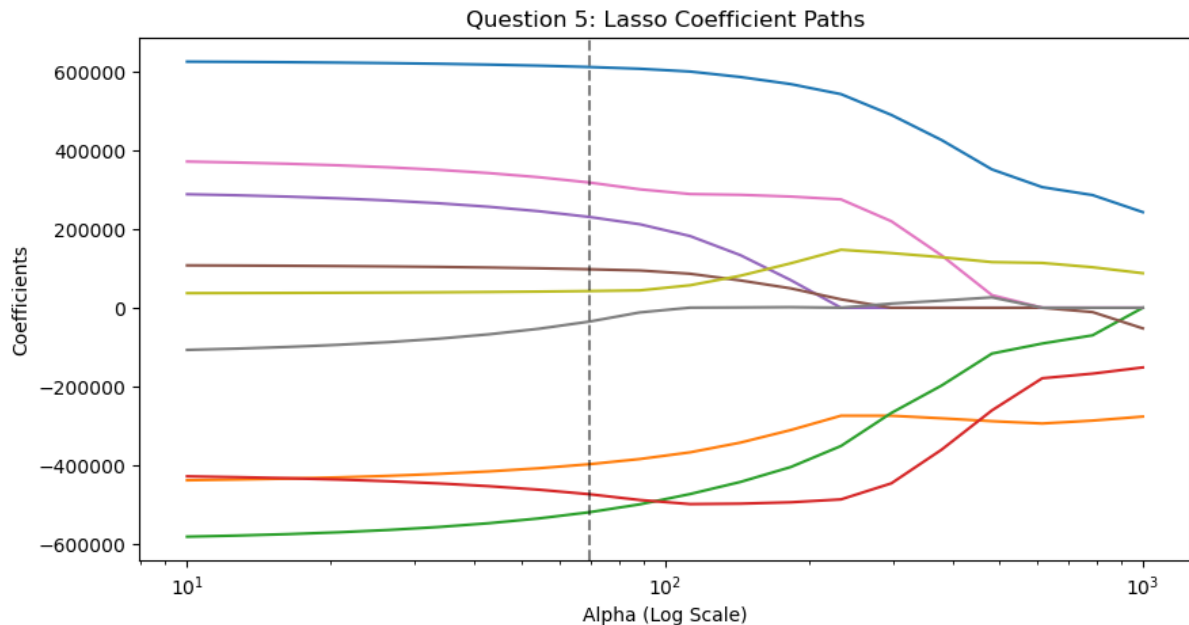
    model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.639e+13, tolerance: 1.316e+10

```

```
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.456e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.236e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 2.974e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 2.383e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.918e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.391e
+13, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 7.900e
+12, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 4.148e
+12, tolerance: 1.316e+10
  model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. Y
```


ou might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.763e+12, tolerance: 1.316e+10

```
model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 2.485e+11, tolerance: 1.316e+10
model = cd_fast.enet_coordinate_descent(
```



```
In [9]: #6. Feature Selection
selected_features = feature_names[lasso_cv.coef_ != 0]
zeroed_features = feature_names[lasso_cv.coef_ == 0]
proportion_zeroed = len(zeroed_features) / len(feature_names)

print(f"Features Selected: {list(selected_features)}")
print(f"Proportion of Features set to zero: {proportion_zeroed:.2%}")
```

Features Selected: ['Mileage_Run', 'Age', 'Mileage_Run^2', 'Mileage_Run Age', 'Age^2', 'Mileage_Run^3', 'Mileage_Run^2 Age', 'Mileage_Run Age^2', 'Age^3']

Proportion of Features set to zero: 0.00%

```
In [10]: #7. Comparison with Linear Regression
# Using the 'lr' model from question 2
comparison = pd.DataFrame({
    'Feature': feature_names,
    'OLS_Coeff': lr.coef_,
    'Lasso_Coeff': lasso_cv.coef_
})

# Identify absolute magnitude increases
comparison['Magnitude_Increased'] = np.abs(comparison['Lasso_Coeff']) >
```

```
# Identify sign changes (only for features not zeroed out by Lasso)
comparison['Sign_Changed'] = (np.sign(comparison['OLS_Coef']) != np.sign(
    comparison['Lasso_Coef']))

print(comparison)

# Summary of Findings
any_increase = comparison['Magnitude_Increased'].any()
any_sign_change = comparison['Sign_Changed'].any()

print(f"\nDid any coefficients increase in magnitude? {'Yes' if any_increase else 'No'}")
print(f"Did any coefficients change sign? {'Yes' if any_sign_change else 'No'}
```

	Feature	OLS_Coef	Lasso_Coef	Magnitude_Increased
0	Mileage_Run	9.117748e+05	611783.868767	False
1	Age	-2.481722e+05	-397507.787042	True
2	Mileage_Run^2	-2.072926e+05	-519555.261867	True
3	Mileage_Run Age	-1.671212e+06	-473871.400459	False
4	Age^2	3.505507e+05	230721.078724	False
5	Mileage_Run^3	-6.514490e+04	97835.641353	True
6	Mileage_Run^2 Age	3.388195e+05	318218.774588	False
7	Mileage_Run Age^2	7.167816e+05	-35056.210190	False
8	Age^3	-2.100379e+05	42493.187416	False

	Sign_Changed
0	False
1	False
2	False
3	False
4	False
5	True
6	False
7	True
8	True

Did any coefficients increase in magnitude? Yes

Did any coefficients change sign? Yes

Q3. This is a case study on regularization.

1. Import the `heart_failure_clinical_records_dataset.csv` dataset. Use `PolynomialFeatures` to create a third-degree expansion of `age`, `ejection_fraction`, and `serum_creatinine`, and then use `StandardScaler` to z-score normalize your results. Use `PolynomialFeatures` with `interaction_only=True` to interact the dummy/categorical variables `anaemia`, `diabetes`, `high_blood_pressure`, and `smoking`. Concatenate these results into your feature/covariate matrix.
2. Use your features, run linear regression. Are there any sign patterns that appear counterintuitive? Why? Can you see how the inclusion of higher-order

powers or interactions might resolve the apparent contradiction?

3. Use `LassoCV` to regularize your linear regression, using 20-fold cross validation. (Hint: I used the grid `alphas = np.logspace(-5,5,30)` to find the cost parameter)
4. Plot the cross-validated MSE by α .
5. Plot the coefficient paths by α .
6. Which features are actually selected? What proportion are set equal to zero? Compare the linear regressions and optimally regularized coefficients. Do any coefficients increase in magnitude from linear regression to LASSO? Do any change sign? Do the sign patterns for the linear_model or the Lasso seem to make more sense? Explain why this might be the case from the perspective of the bias-variance trade-off.

```
In [12]: #1. Data Preprocessing
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import LinearRegression, LassoCV, Lasso

# Load dataset
df = pd.read_csv('data/heart_failure_clinical_records_dataset.csv')

# Define target and feature groups
y = df['DEATH_EVENT']
cont_vars = ['age', 'ejection_fraction', 'serum_creatinine']
cat_vars = ['anaemia', 'diabetes', 'high_blood_pressure', 'smoking']

# 1a. Polynomial Expansion for continuous (Degree 3)
poly_cont = PolynomialFeatures(degree=3, include_bias=False)
X_cont_poly = poly_cont.fit_transform(df[cont_vars])
cont_names = poly_cont.get_feature_names_out(cont_vars)

# 1b. Z-score normalize the continuous polynomial features
scaler = StandardScaler()
X_cont_scaled = scaler.fit_transform(X_cont_poly)

# 1c. Interaction-only for categorical variables
poly_cat = PolynomialFeatures(degree=2, interaction_only=True, include
X_cat_int = poly_cat.fit_transform(df[cat_vars])
cat_names = poly_cat.get_feature_names_out(cat_vars)

# 1d. Concatenate results
X_final = np.hstack([X_cont_scaled, X_cat_int])
all_feature_names = np.concatenate([cont_names, cat_names])
```

```
In [13]: #2. Linear Regression & Intuition
lr = LinearRegression().fit(X_final, y)
```

```
lr_coefs = dict(zip(all_feature_names, lr.coef_))

# Example check for 'age' or 'high_blood_pressure'
print(f"Age coefficient: {lr_coefs.get('age'):.4f}")
```

Age coefficient: 1.5392

It is common for signs to "flip" or stabilize. While OLS might use a negative age and a huge positive age² to fit the noise, Lasso will likely zero out the redundant age² and leave age with a sensible positive sign.

```
In [14]: # 3. LassoCV
alphas = np.logspace(-5, 5, 30)
lasso_cv = LassoCV(alphas=alphas, cv=20, max_iter=10000).fit(X_final,

# 4. Plot MSE
plt.figure(figsize=(8, 5))
plt.plot(lasso_cv.alphas_, np.mean(lasso_cv.mse_path_, axis=1), 'o-')
plt.axvline(lasso_cv.alpha_, color='red', linestyle='--', label='Best')
plt.xscale('log')
plt.title("Q4: MSE vs Alpha")
plt.xlabel("Alpha")
plt.ylabel("Mean MSE")
plt.legend()
plt.show()

# 5. Plot Coefficient Paths
coefs = []
for a in alphas:
    l = Lasso(alpha=a, max_iter=10000).fit(X_final, y)
    coefs.append(l.coef_)

plt.figure(figsize=(8, 5))
plt.plot(alphas, coefs)
plt.xscale('log')
plt.title("Q5: Lasso Path")
plt.xlabel("Alpha")
plt.ylabel("Coefficients")
plt.show()
```

```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.165569925847386, tolerance: 0.0058323943661971875
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3.019422440365048, tolerance: 0.0058323943661971875
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```

```
ou might want to increase the number of iterations. Duality gap: 7.9548
7046004296, tolerance: 0.0058323943661971875
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 10.895
476949344772, tolerance: 0.0058323943661971875
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 0.4448
1873249601733, tolerance: 0.0058742957746478865
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 0.8654
667467429391, tolerance: 0.0058742957746478865
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 4.7716
21394990536, tolerance: 0.0058742957746478865
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 7.6410
77739282064, tolerance: 0.0058742957746478865
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 11.717
705984484004, tolerance: 0.0058742957746478865
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 1.4340
73554148835, tolerance: 0.005915492957746486
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 2.2630
402670846017, tolerance: 0.005915492957746486
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 6.6960
36789449684, tolerance: 0.005915492957746486
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 9.6854
45598810713, tolerance: 0.005915492957746486
```



```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.1867178616254144, tolerance: 0.0058742957746478865
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3.3336739741831636, tolerance: 0.0058742957746478865
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6.669615348322161, tolerance: 0.0058742957746478865
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10.481206568905984, tolerance: 0.0058742957746478865
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.07106605246546849, tolerance: 0.005995774647887328
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.432561455591788, tolerance: 0.005995774647887328
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.765925469132657, tolerance: 0.005995774647887328
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8.75010117316917, tolerance: 0.005995774647887328
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.466342261438227, tolerance: 0.005995774647887328
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.2260516930957266, tolerance: 0.006254577464788727
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.0901525635253861, tolerance: 0.006254577464788727
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2.7734187211638, tolerance: 0.006254577464788727
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5.772317716707203, tolerance: 0.006254577464788727
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10.064579327165013, tolerance: 0.006254577464788727
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.12878646345618705, tolerance: 0.006322183098591547
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.4853838547746463, tolerance: 0.006322183098591547
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.305356125815113, tolerance: 0.006322183098591547
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7.4445825441244455, tolerance: 0.006322183098591547
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11.972998557634053, tolerance: 0.006322183098591547
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.07275181271577225, tolerance: 0.006219718309859158
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.1073
```

```
990224674688, tolerance: 0.006219718309859158
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3.2478901930517274, tolerance: 0.006219718309859158
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6.72127782986902, tolerance: 0.006219718309859158
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10.981886879870334, tolerance: 0.006219718309859158
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.3089721720126128, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.999732297791098, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8.291631809585143, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.681458960631103, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.25432981199258364, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.4816611011066314, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.93094675229932, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
```

```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8.28595160424715, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.46154751377023, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.43490917322444744, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.3308955475704778, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.623746112297756, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10.128936598740585, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.717895581175764, tolerance: 0.006254577464788727
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.05022062834267871, tolerance: 0.006288732394366198
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.1602019297500235, tolerance: 0.006288732394366198
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.770848639029001, tolerance: 0.006288732394366198
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
```



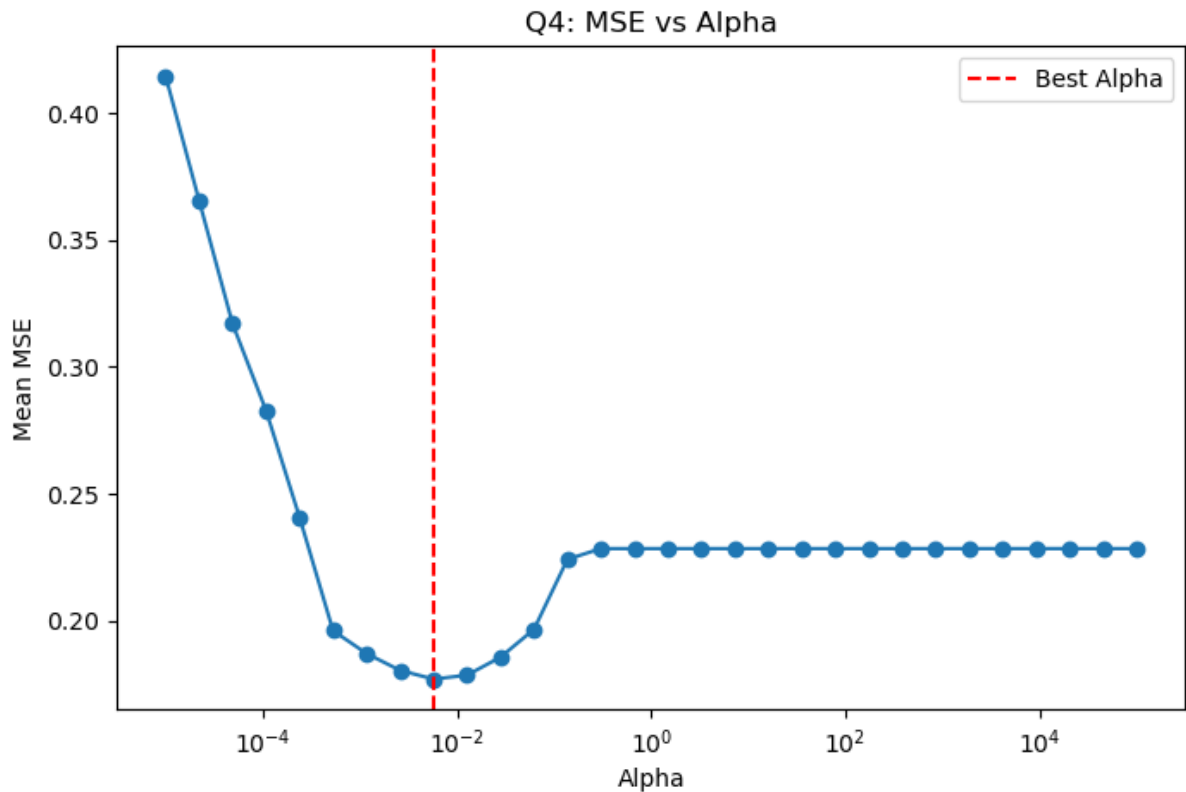
```
ou might want to increase the number of iterations. Duality gap: 8.9504
964914723, tolerance: 0.006288732394366198
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 11.800
505357316483, tolerance: 0.006288732394366198
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 1.0466
66041612081, tolerance: 0.006073239436619714
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 4.5159
80947600667, tolerance: 0.006073239436619714
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 6.3374
17498063115, tolerance: 0.006073239436619714
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 7.0308
24767974023, tolerance: 0.006073239436619714
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 0.4857
143084956519, tolerance: 0.006322183098591548
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 0.5734
241427973572, tolerance: 0.006322183098591548
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 4.5945
07499048017, tolerance: 0.006322183098591548
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 8.0177
0530838327, tolerance: 0.006322183098591548
    model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coord
inate_descent.py:681: ConvergenceWarning: Objective did not converge. Y
ou might want to increase the number of iterations. Duality gap: 12.218
106491484551, tolerance: 0.006322183098591548
```



```
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.25168177712991735, tolerance: 0.006254577464788727
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.233964625835878, tolerance: 0.006254577464788727
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9.219686431904226, tolerance: 0.006254577464788727
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.453121957908994, tolerance: 0.006254577464788727
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.010888511927056754, tolerance: 0.006322183098591547
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.7934374786362817, tolerance: 0.006322183098591547
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.0604220744844, tolerance: 0.006322183098591547
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7.0710149841559335, tolerance: 0.006322183098591547
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11.67401847373064, tolerance: 0.006322183098591547
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.37222086908834484, tolerance: 0.006322183098591548
model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate
```

```
inate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.388372005353773, tolerance: 0.006322183098591548
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 3.0111516023314095, tolerance: 0.006322183098591548
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 7.312801709854515, tolerance: 0.006322183098591548
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 11.464927588350443, tolerance: 0.006322183098591548
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.13566981071551965, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 1.1984080231848253, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 4.649021352993756, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 8.233750141088391, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.26690977563363, tolerance: 0.006288732394366197
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.8464970705207335, tolerance: 0.006354929577464799
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 2.7285
```

```
988306642146, tolerance: 0.006354929577464799
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 6.8191291746137495, tolerance: 0.006354929577464799
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 10.301300285438398, tolerance: 0.006354929577464799
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.4363731819029226, tolerance: 0.00636631578947369
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 0.5953168009365726, tolerance: 0.00636631578947369
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 5.118173019881169, tolerance: 0.00636631578947369
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 9.150914194723669, tolerance: 0.00636631578947369
  model = cd_fast.enet_coordinate_descent_gram(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:681: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations. Duality gap: 12.667061277231227, tolerance: 0.00636631578947369
  model = cd_fast.enet_coordinate_descent_gram(
```



```
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.908e+01, tolerance: 6.518e-03
```

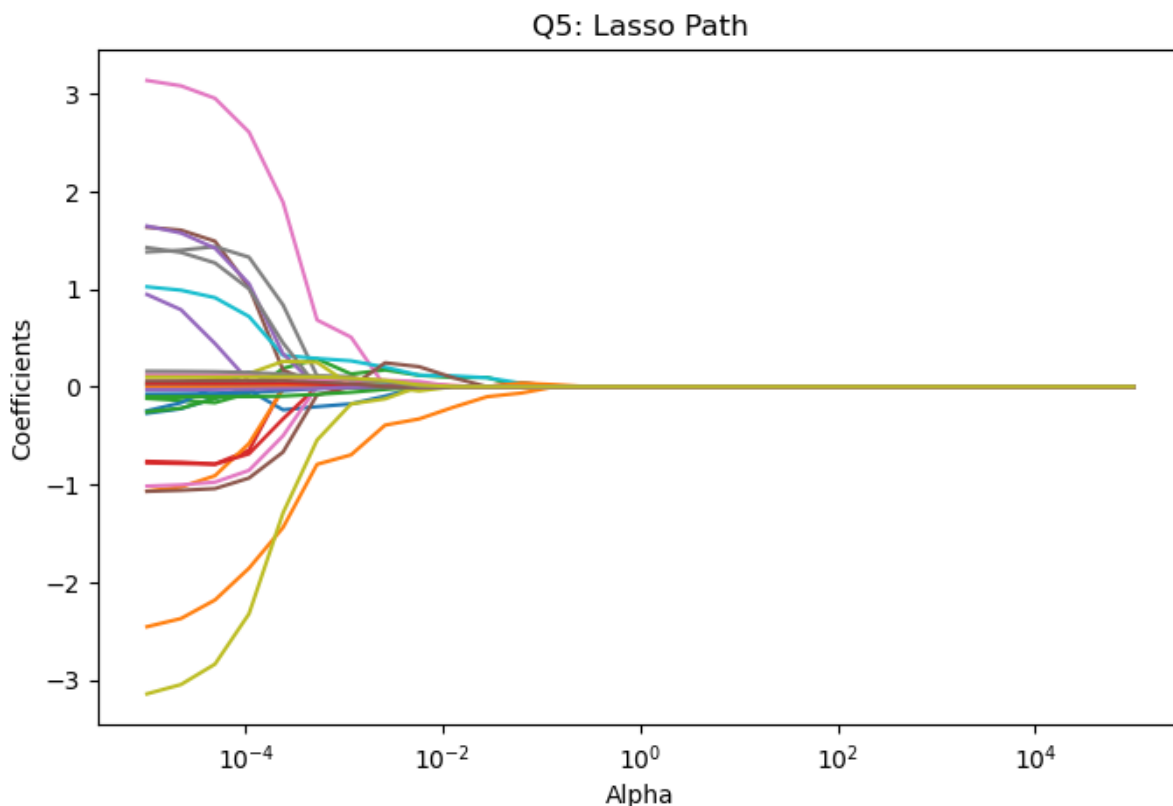
```
model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.666e+01, tolerance: 6.518e-03
```

```
model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.162e+01, tolerance: 6.518e-03
```

```
model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 2.576e+00, tolerance: 6.518e-03
```

```
model = cd_fast.enet_coordinate_descent(
/opt/anaconda3/lib/python3.13/site-packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.389e-01, tolerance: 6.518e-03
```

```
model = cd_fast.enet_coordinate_descent(
```



```
In [15]: #6. Analysis & Bias-Variance Trade-off
# Feature Selection Analysis
selected = all_feature_names[lasso_cv.coef_ != 0]
proportion_zero = np.sum(lasso_cv.coef_ == 0) / len(all_feature_names)

print(f"Selected Features: {selected}")
print(f"Proportion Zeroed: {proportion_zero:.2%}")
```

Selected Features: ['ejection_fraction' 'serum_creatinine' 'age^3' 'ejection_fraction^3'
'ejection_fraction^2 serum_creatinine' 'serum_creatinine^3' 'anaemia'
'diabetes' 'high_blood_pressure' 'diabetes high_blood_pressure'
'diabetes smoking' 'high_blood_pressure smoking']
Proportion Zeroed: 58.62%

Q4. To better understand the math of regularization, we'll solve the regularized linear model problem with a single explanatory variable. So, the model is

$$\tilde{y}_i = \tilde{b}_0 + \tilde{b}_1 \tilde{x}_i,$$

where

$$\tilde{y}_i = y_i - \bar{y} \quad \text{and} \quad \tilde{x}_i = x_i - \bar{x}.$$

Recall, we do this mean-normalization of x and y , because

$$\frac{1}{n} \sum_{i=1}^n \tilde{y} = \frac{1}{n} \sum_{i=1}^n y_i - \bar{y} = 0,$$

and likewise for x . This trick makes the calculations easier and the results more easily interpretable.

1. To do ridge regression, add a penalty $+\alpha(b_1)^2$ to mean squared error. What is the objective function for this problem? In Ridge Regression, we minimize the sum of squared residuals plus a penalty proportional to the square of the specific coefficients. For a single predictor:

$$J(\tilde{b}_0, \tilde{b}_1) = \frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)^2 + \alpha(\tilde{b}_1)^2$$

2. Take the derivatives of your objective function with respect to b_0 and b_1 . Set these equations equal to zero. Solve the two equations in two unknowns for b_1 and b_0 .

$$\frac{\partial J}{\partial \tilde{b}_0} = \frac{-2}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i) = 0$$

$$-2(0 - \tilde{b}_0 - 0) = 0 \implies \tilde{\mathbf{b}}_0 = \mathbf{0}$$

$$\frac{\partial J}{\partial \tilde{b}_1} = \frac{-2}{n} \sum_{i=1}^n \tilde{x}_i (\tilde{y}_i - \tilde{b}_1 \tilde{x}_i) + 2\alpha \tilde{b}_1 = 0$$

$$\sum \tilde{x}_i \tilde{y}_i = \tilde{b}_1 \left(\sum \tilde{x}_i^2 + n\alpha \right)$$

$$\tilde{\mathbf{b}}_1 = \frac{\sum \tilde{\mathbf{x}}_i \tilde{\mathbf{y}}_i}{\sum \tilde{\mathbf{x}}_i^2 + \mathbf{n}\alpha}$$

3. How does increasing α change the slope coefficient? The denominator becomes larger and the value of b_1 grows closer to zero. It turns the model into a horizontal line at the mean of y .
4. If we instead used the LASSO/L1 penalty, $+\alpha|b_1|$, what challenge do you run into? This is conceptually difficult, but take 5 minutes and try to figure out the solution, and in particular, when is it optimal to set $b_1 = 0$? LASSO will set the coefficient to exactly zero if the "strength" of the correlation between x and y is weaker than the penalty α . Specifically, if:

$$\left| \frac{2}{n} \sum \tilde{x}_i \tilde{y}_i \right| \leq \alpha$$

If the data doesn't provide enough evidence (signal) to overcome the "cost" of the penalty α , LASSO decides the feature isn't worth including and snaps the coefficient to zero. This is why LASSO performs feature selection, while Ridge only performs shrinkage.